

Restaurant Data Analysis

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
import os

os.listdir()

['.config', 'sample_data']
```

Load Dataset

```
from google.colab import files

uploaded = files.upload()
```

No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
import os

print(os.getcwd())
print(os.listdir())

/content
['.config', 'Dataset .csv', 'sample_data']
```

```
df = pd.read_csv("Dataset .csv")
```

```
import os
print(os.listdir())

['.config', 'Dataset .csv', 'sample_data']
```

```
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Current
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswa Pula(
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswa Pula(
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswa Pula(
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswa Pula(
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswa Pula(

5 rows × 21 columns

Data Understanding

```
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Current
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswa Pula(
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswa Pula(
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswa Pula(
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswa Pula(
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswa Pula(

5 rows × 21 columns

df.tail()

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines
9546	5915730	Namlı Gurme	208	İstanbul	Kemankeş Karamustafa Paşa Mahallesi, Rıhtım ...	Karaköy	Karaköy, İstanbul	28.977392	41.022793	Turkish
9547	5908749	Ceviz Aca	208	İstanbul	Koşuyolu Mahallesi, Muhtitin İstanbul Caddesi	Koşuyolu	Koşuyolu, İstanbul	29.041297	41.009847	World Cuisine Patisserie Cafe
9548	5915807	Huqqa	208	İstanbul	Kuruçeşme Mahallesi, Muallim Naci Caddesi, N...	Kuruçeşme	Kuruçeşme, İstanbul	29.034640	41.055817	Italian World Cuisine
9549	5916112	Ak Kahve	208	İstanbul	Kuruçeşme Mahallesi, Muallim Naci Caddesi, N...	Kuruçeşme	Kuruçeşme, İstanbul	29.036019	41.057979	Restaurant Cafe
9550	5927402	Walter's Coffee Roastery	208	İstanbul	Cafea Pa Mahallesi, Bademaltı Sokak, No 21/B, ...	Moda	Moda, İstanbul	29.026016	40.984776	Cafe

5 rows × 21 columns

df.shape

(9551, 21)

df.columns

```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
      'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
      'Average Cost for two', 'Currency', 'Has Table booking',
      'Has Online delivery', 'Is delivering now', 'Switch to order menu',
      'Price range', 'Aggregate rating', 'Rating color', 'Rating text',
      'Votes'],
      dtype='object')
```

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Restaurant ID                        9551 non-null   int64
1   Restaurant Name                      9551 non-null   object
2   Country Code                        9551 non-null   int64
3   City                                9551 non-null   object
4   Address                             9551 non-null   object
5   Locality                            9551 non-null   object
6   Locality Verbose                    9551 non-null   object
7   Longitude                           9551 non-null   float64
8   Latitude                           9551 non-null   float64
9   Cuisines                            9542 non-null   object
10  Average Cost for two                 9551 non-null   int64
11  Currency                            9551 non-null   object
12  Has Table booking                    9551 non-null   object
13  Has Online delivery                  9551 non-null   object
14  Is delivering now                    9551 non-null   object
15  Switch to order menu                 9551 non-null   object
16  Price range                         9551 non-null   int64
17  Aggregate rating                     9551 non-null   float64
18  Rating color                        9551 non-null   object
19  Rating text                         9551 non-null   object
20  Votes                               9551 non-null   int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

df.describe()

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	Aggregate rating	Votes
count	9.551000e+03	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000
mean	9.051128e+06	18.365616	64.126574	25.854381	1199.210763	1.804837	2.666370	156.909748
std	8.791521e+06	56.750546	41.467058	11.007935	16121.183073	0.905609	1.516378	430.169145
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	0.000000	0.000000
25%	3.019625e+05	1.000000	77.081343	28.478713	250.000000	1.000000	2.500000	5.000000
50%	6.004089e+06	1.000000	77.191964	28.570469	400.000000	2.000000	3.200000	31.000000
75%	1.835229e+07	1.000000	77.282006	28.642758	700.000000	2.000000	3.700000	131.000000
max	1.855000e+07	1.000000	77.282006	28.642758	700.000000	2.000000	3.700000	131.000000

Data Cleaning

df.isnull().sum()

	0
Restaurant ID	0
Restaurant Name	0
Country Code	0
City	0
Address	0
Locality	0
Locality Verbose	0
Longitude	0
Latitude	0
Cuisines	9
Average Cost for two	0
Currency	0
Has Table booking	0
Has Online delivery	0
Is delivering now	0
Switch to order menu	0
Price range	0
Aggregate rating	0
Rating color	0
Rating text	0
Votes	0

dtype: int64

df.duplicated().sum()

np.int64(0)

df = df.drop_duplicates()

df.dtypes

	0
Restaurant ID	int64
Restaurant Name	object
Country Code	int64
City	object
Address	object
Locality	object
Locality Verbose	object
Longitude	float64
Latitude	float64
Cuisines	object
Average Cost for two	int64
Currency	object
Has Table booking	object
Has Online delivery	object
Is delivering now	object
Switch to order menu	object
Price range	int64
Aggregate rating	float64
Rating color	object
Rating text	object
Votes	int64

dtype: object

```
df["Votes"] = df["Votes"].astype(int)
```

```
df["City"].unique()
```

```
array(['Makati City', 'Mandaluyong City', 'Pasay City', 'Pasig City',
      'Quezon City', 'San Juan City', 'Santa Rosa', 'Tagaytay City',
      'Taguig City', 'Brasilia', 'Rio de Janeiro', 'Sao Paulo',
      'Albany', 'Armidale', 'Athens', 'Augusta', 'Balingup',
      'Beechworth', 'Boise', 'Cedar Rapids/Iowa City', 'Chatham-Kent',
      'Clatskanie', 'Cochrane', 'Columbus', 'Consort', 'Dalton',
      'Davenport', 'Des Moines', 'Dickty Beach', 'Dubuque',
      'East Ballina', 'Fernley', 'Flaxton', 'Forrest', 'Gainesville',
      'Hepburn Springs', 'Huskisson', 'Inverloch', 'Lakes Entrance',
      'Lakeview', 'Lincoln', 'Lorn', 'Macedon', 'Macon', 'Mayfield',
      'Mc Millan', 'Middleton Beach', 'Miller', 'Monroe', 'Montville',
      'Ojo Caliente', 'Orlando', 'Palm Cove', 'Paynesville', 'Penola',
      'Pensacola', 'Phillip Island', 'Pocatello', 'Potrero', 'Princeton',
      'Rest of Hawaii', 'Savannah', 'Singapore', 'Sioux City',
      'Tampa Bay', 'Tanunda', 'Trentham East', 'Valdosta', 'Vernonia',
      'Victor Harbor', 'Vineland Station', 'Waterloo', 'Weirton',
      'Winchester Bay', 'Yorkton', 'Abu Dhabi', 'Dubai', 'Sharjah',
      'Agra', 'Ahmedabad', 'Allahabad', 'Amritsar', 'Aurangabad',
      'Bangalore', 'Bhopal', 'Bhubaneshwar', 'Chandigarh', 'Chennai',
      'Coimbatore', 'Dehradun', 'Faridabad', 'Ghaziabad', 'Goa',
      'Gurgaon', 'Guwahati', 'Hyderabad', 'Indore', 'Jaipur', 'Kanpur',
      'Kochi', 'Kolkata', 'Lucknow', 'Ludhiana', 'Mangalore', 'Mohali',
      'Mumbai', 'Mysore', 'Nagpur', 'Nashik', 'New Delhi', 'Noida',
      'Panchkula', 'Patna', 'Puducherry', 'Pune', 'Ranchi',
      'Secunderabad', 'Surat', 'Vadodara', 'Varanasi', 'Vizag',
      'Bandung', 'Bogor', 'Jakarta', 'Tangerang', 'Auckland',
      'Wellington City', 'Birmingham', 'Edinburgh', 'London',
      'Manchester', 'Doha', 'Cape Town', 'Inner City', 'Johannesburg',
      'Pretoria', 'Randburg', 'Sandton', 'Colombo', 'Ankara',
      'Istanbul'], dtype=object)
```

```
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Current
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswa Pula(
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2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswa Pula(
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswa Pula(
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswa Pula(

5 rows × 21 columns

```
df[df['Cuisines'].isnull()]
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency
84	17284105	Cookie Shoppe	216	Albany	115 N Jackson St, Albany, GA 31701	Albany	Albany, Albany	-84.154000	31.577200	NaN	...	Dollar(\$)
87	17284211	Pearly's Famous Country Cookng	216	Albany	814 N Slappeg Blvd, Albany, GA 31701	Albany	Albany, Albany	-84.175900	31.588200	NaN	...	Dollar(\$)
94	17284158	Jimmie's Hot Dogs	216	Albany	204 S Jackson St, Albany, GA 31701	Albany	Albany, Albany	-84.153400	31.575100	NaN	...	Dollar(\$)
297	17374552	Corkscrew Cafe	216	Gainesville	51 W Main St, Dahlonaga, GA 30533	Dahlonaga	Dahlonaga, Gainesville	-83.985800	34.531800	NaN	...	Dollar(\$)
328	17501439	Dovetail	216	Macon	543 Cherry St, Macon, GA 31201	Macon	Macon, Macon	-83.627979	32.836410	NaN	...	Dollar(\$)
346	17606621	HI Lite Bar & Lounge	216	Miller	109 N Broadway Ave, Miller, SD 57362	Miller	Miller, Miller	-98.989100	44.515800	NaN	...	Dollar(\$)
368	17059060	Hillstone	216	Orlando	215 South Orlando Avenue, Winter Park, FL 32789	Winter Park	Winter Park, Orlando	-81.365260	28.596682	NaN	...	Dollar(\$)
418	17142698	Leonard's Bakery	216	Rest of Hawaii	933 Kapahulu Ave, Honolulu, HI 96816	Kaimuki	Kaimuki, Rest of Hawaii	-157.813432	21.284586	NaN	...	Dollar(\$)
455	17616465	Tybee Island Social Club	216	Savannah	1311 Butler Ave, Tybee Island, GA 31328	Tybee Island	Tybee Island, Savannah	-80.848297	31.995810	NaN	...	Dollar(\$)

9 rows × 21 columns

```
df = df.dropna(subset=['Cuisines'])
```

```
df.describe()
```

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	Aggregate rating	Votes
count	9.542000e+03	9542.000000	9542.000000	9542.000000	9542.000000	9542.000000	9542.000000	9542.000000
mean	9.043301e+06	18.179208	64.274997	25.848532	1200.326137	1.804968	2.665238	156.772060
std	8.791967e+06	56.451600	41.197602	11.010094	16128.743876	0.905563	1.516588	430.203324
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	0.000000	0.000000
25%	3.019312e+05	1.000000	77.081565	28.478658	250.000000	1.000000	2.500000	5.000000
50%	6.002726e+06	1.000000	77.192031	28.570444	400.000000	2.000000	3.200000	31.000000
75%	1.835260e+07	1.000000	77.282043	28.642711	700.000000	2.000000	3.700000	130.000000

```
df['Cuisines'].head(10)
```

```

      Cuisines
0    French, Japanese, Desserts
1                Japanese
2    Seafood, Asian, Filipino, Indian
3                Japanese, Sushi
4                Japanese, Korean
5                Chinese
6                Asian, European
7    Seafood, Filipino, Asian, European
8                European, Asian, Indian
9                Filipino

```

dtype: object

```

cuisines = df['Cuisines'].str.split(' ', ' ')
cuisines = cuisines.explode()

```

```
cuisines.head(15)
```

```

      Cuisines
0    French
0    Japanese
0    Desserts
1    Japanese
2    Seafood
2    Asian
2    Filipino
2    Indian
3    Japanese
3    Sushi
4    Japanese
4    Korean
5    Chinese
6    Asian
6    European

```

dtype: object

```

cuisine_counts = cuisines.value_counts()

cuisine_counts

```

	count
Cuisines	
North Indian	3960
Chinese	2735
Fast Food	1986
Mughlai	995
Italian	764
...	...
Cuisine Varies	1
Fish and Chips	1
Durban	1
D☐_ner	1
B☐_rek	1

145 rows × 1 columns

dtype: int64

cuisine_counts

	count
Cuisines	
North Indian	3960
Chinese	2735
Fast Food	1986
Mughlai	995
Italian	764
...	...
Cuisine Varies	1
Fish and Chips	1
Durban	1
D☐_ner	1
B☐_rek	1

145 rows × 1 columns

dtype: int64

top3 = cuisine_counts.head(3)

top3

	count
Cuisines	
North Indian	3960
Chinese	2735
Fast Food	1986

dtype: int64

total_cuisines = cuisine_counts.sum()

percentage = (top3 / total_cuisines) * 100

percentage.round(2)

	count
North Indian	20.09
Chinese	13.88
Fast Food	10.08

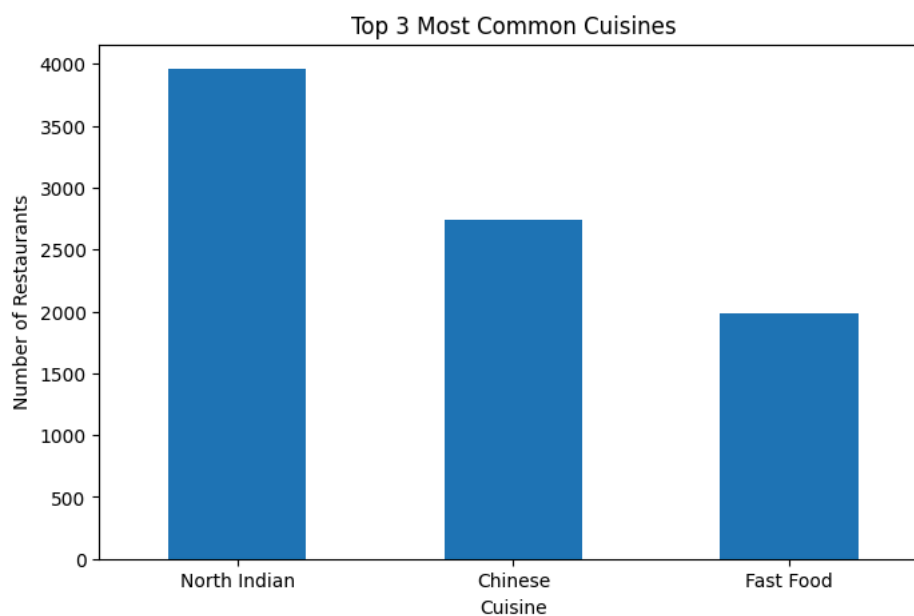
dtype: float64

```
plt.figure(figsize=(8,5))

top3.plot(kind='bar')

plt.title("Top 3 Most Common Cuisines")
plt.xlabel("Cuisine")
plt.ylabel("Number of Restaurants")

plt.xticks(rotation=0) # Cuisine names seedhe dikhne ke liye
plt.show()
```



✓ Level 1 - Task 1: Top Cuisines Analysis

✓ Objective

Identify the top three most common cuisines and calculate their percentage distribution.

```
cuisines = df['Cuisines'].str.split(', ').explode()

cuisines.head(10)
```

Cuisines

```
0    French
0    Japanese
0    Desserts
1    Japanese
2    Seafood
2    Asian
2    Filipino
2    Indian
3    Japanese
3    Sushi
```

dtype: object

```
cuisine_counts = cuisines.value_counts()
```

```
cuisine_counts.head(10)
```

	count
North Indian	3960
Chinese	2735
Fast Food	1986
Mughlai	995
Italian	764
Bakery	745
Continental	736
Cafe	703
Desserts	653
South Indian	636

dtype: int64

```
top3 = cuisine_counts.head(3)
```

```
top3
```

	count
North Indian	3960
Chinese	2735
Fast Food	1986

dtype: int64

```
total_cuisines = cuisine_counts.sum()
```

```
percentage = (top3 / total_cuisines) * 100
```

```
percentage.round(2)
```

	count
Cuisines	
North Indian	20.09
Chinese	13.88
Fast Food	10.08

dtype: float64

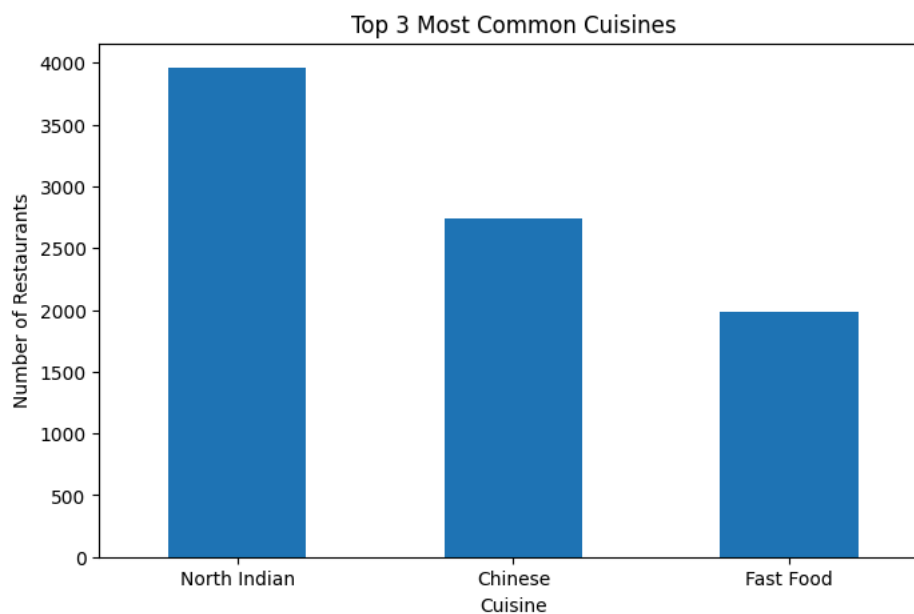
```
plt.figure(figsize=(8,5))

top3.plot(kind='bar')

plt.title("Top 3 Most Common Cuisines")
plt.xlabel("Cuisine")
plt.ylabel("Number of Restaurants")

plt.xticks(rotation=0)

plt.show()
```



Conclusion

The analysis shows that North Indian, Chinese, and Fast Food are the three most popular cuisines in the dataset. Among them, North Indian cuisine has the highest number of restaurants, indicating its popularity.

Level 1 - Task 1: Top Cuisines Analysis ← Text Cell

Objective ← Text Cell

- Code 1 cuisines = df['Cuisines'].str.split(',').explode()
- Code 2 cuisine_counts = cuisines.value_counts() cuisine_counts.head(10)
- Code 3 top3 = cuisine_counts.head(3) top3
- Code 4 total_cuisines = cuisine_counts.sum() percentage = (top3 / total_cuisines) * 100 percentage.round(2)
- Code 5 Bar Chart

Observation ← Text Cell

Conclusion ← Text Cell

✓ Level 1 - Task 2: City Analysis

✓ Objective

Identify the city with the highest number of restaurants and calculate the average rating for each city.

```
city_counts = df['City'].value_counts()  
city_counts.head(10)
```

	count
City	
New Delhi	5473
Gurgaon	1118
Noida	1080
Faridabad	251
Ghaziabad	25
Bhubaneswar	21
Ahmedabad	21
Lucknow	21
Guwahati	21
Amritsar	21

dtype: int64

```
top_city = city_counts.head(1)  
top_city
```

	count
City	
New Delhi	5473

dtype: int64

```
city_rating = df.groupby('City')['Aggregate rating'].mean()  
city_rating.head(10)
```

Aggregate rating	
City	
Abu Dhabi	4.300000
Agra	3.965000
Ahmedabad	4.161905
Albany	3.552941
Allahabad	3.395000
Amritsar	3.685714
Ankara	4.305000
Armidale	3.500000
Athens	4.200000
Auckland	4.275000

dtype: float64

```
city_rating = df.groupby('City')['Aggregate rating'].mean()

city_rating.head(10)
```

Aggregate rating	
City	
Abu Dhabi	4.300000
Agra	3.965000
Ahmedabad	4.161905
Albany	3.552941
Allahabad	3.395000
Amritsar	3.685714
Ankara	4.305000
Armidale	3.500000
Athens	4.200000
Auckland	4.275000

dtype: float64

```
highest_rating_city = city_rating.sort_values(ascending=False)

highest_rating_city.head(10)
```

Aggregate rating	
City	
Inner City	4.900000
Quezon City	4.800000
Makati City	4.650000
Pasig City	4.633333
Mandaluyong City	4.625000
Beechworth	4.600000
London	4.535000
Taguig City	4.525000
Lincoln	4.500000
Tagaytay City	4.500000

dtype: float64

```
highest_rating_city = city_rating.sort_values(ascending=False)
highest_rating_city.head(10)
```

Aggregate rating	
City	
Inner City	4.900000
Quezon City	4.800000
Makati City	4.650000
Pasig City	4.633333
Mandaluyong City	4.625000
Beechworth	4.600000
London	4.535000
Taguig City	4.525000
Lincoln	4.500000
Tagaytay City	4.500000

dtype: float64

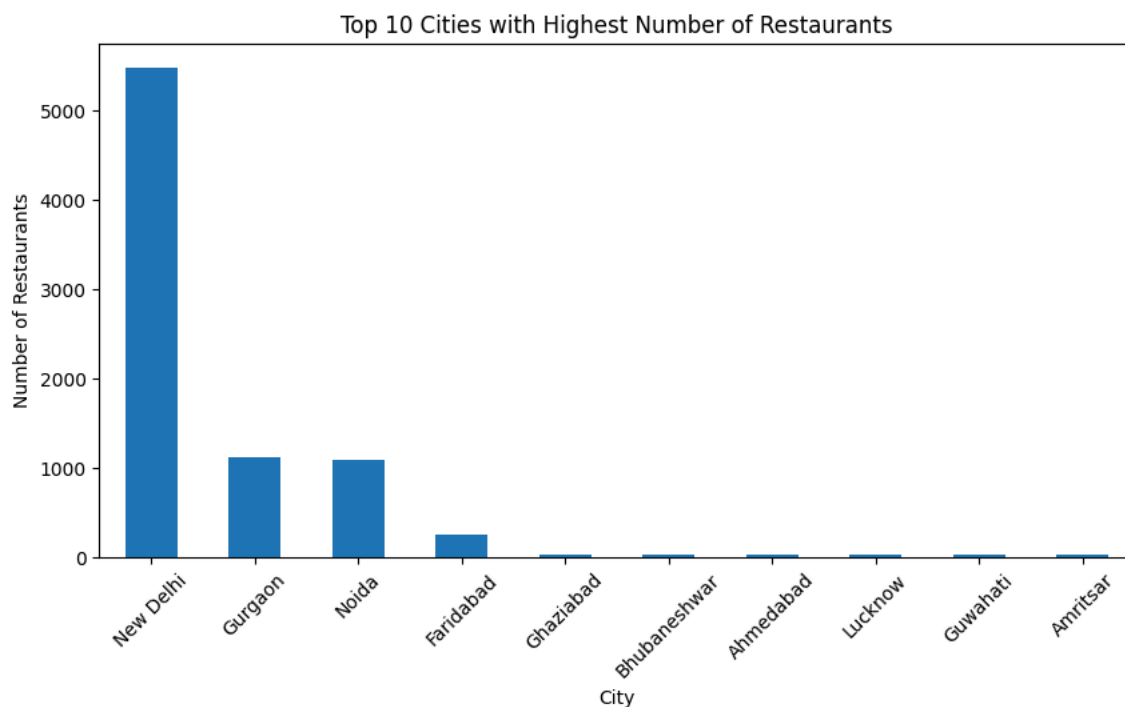
```
plt.figure(figsize=(10,5))

city_counts.head(10).plot(kind='bar')

plt.title("Top 10 Cities with Highest Number of Restaurants")
plt.xlabel("City")
plt.ylabel("Number of Restaurants")

plt.xticks(rotation=45)

plt.show()
```



Observation

- New Delhi has the highest number of restaurants (5473).
- Gurgaon and Noida are the second and third cities with the highest number of restaurants.
- Most other cities have significantly fewer restaurants compared to New Delhi.

- Based on the average rating analysis, Inner City has the highest average restaurant rating of 4.9.

Conclusion

The analysis shows that New Delhi has the largest number of restaurants in the dataset. However, the city with the highest average restaurant rating is Inner City (4.9), indicating that the city with the most restaurants does not necessarily have the highest average ratings.

✓ Level 1 - Task 3: Price Range Analysis

✓ Objective

Analyze the distribution of restaurants across different price ranges and identify the most common price range.

```
price_counts = df['Price range'].value_counts().sort_index()
price_counts
```

	count
Price range	